UNIVERSITI TEKNOLOGI PETRONAS

Artificial Intelligent
LAB-1

BFS

Problem 1: Graph Path Search
Find a path from 'A' to 'G' using BFS on a weighted graph (weights ignored).
Input:

```
graph = {
'A': ['B', 'C'],
'B': ['D', 'E'],
'C': ['F'],
'D': ['G'],
'E': ['G'],
'F': ['G'],
'G': []
}
```

Problem 2: Maze Path Search
Find a path from 'S' to 'G' in a 2D grid using BFS. '.' = open, '#' = wall.
Input:

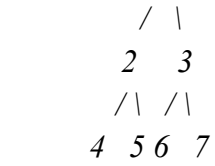
```
maze = [
['S', '.', '.', '#', 'G'],
['#', '#', '.', '#', '.'],
['.', '.', '.', '.', '.'],
['.', '#', '#', '#', '.'],
['.', '.', '.', '.', '.']
]
```

Problem 3: Word Transformation Path
Transform 'hit' to 'cog' by changing one letter at a time, each intermediate word must be valid.
Input:

word_list = ["hit", "hot", "dot", "dog", "lot", "log", "cog"]

Problem 4: Binary Tree Level Order Traversal
Print all nodes level-wise using BFS (Level Order Traversal).
Input:

Binary Tree 1



DFS

Problem 1: Graph Path Search
Task:
Find a path from A to G using DFS.
Graph Representation:

```
graph = {
'A': ['B', 'C'],
'B': ['D', 'E'],
'C': ['F'],
'D': ['G'],
'E': ['G'],
'F': ['G'],
'G': []
}
```

Problem 2: Maze Solver
Task:
Find a path from 'S' to 'G' using DFS in a 2D maze.

```
maze = [
['S', '.', '.', '#', 'G'],
['#', '#', '.', '#', '.'],
['.', '.', '.', '.', '.'],
['.', '#', '#', '#', '.'],
['.', '.', '.', '.', '.']
]
```

Problem 4: Tree Traversal (Preorder)
Task:
Traverse a binary tree using DFS (Preorder).

UCS

Problem 1: Path Cost Search
Graph with Weights:

```
graph = {
   'A': [('B', 1), ('C', 4)],
   'B': [('D', 3), ('E', 2)],
   'C': [('F', 5)],
   'D': [('G', 4)],
   'E': [('G', 1)],
   'F': [('G', 2)],
   'G': []
}
```

Problem 2: Grid with Costs

```
grid = [
   [1, 1, 1, 99, 1],
   [99, 99, 1, 99, 1],
   [1, 1, 1, 1, 1],
   [1, 99, 99, 99, 1],
   [1, 1, 1, 1, 1]
]
```

Problem 3: Graph with Loops

```
graph = {
   'A': [('B', 2), ('C', 5)],
   'B': [('A', 2), ('D', 1)],
```

```
    'C': [('A', 5), ('D', 2)],
    'D': [('B', 1), ('C', 2), ('G', 3)],
    'G': []
}
```
